

A Reproducible Mobility Equity Score: Integrating Transit, Walkability, Bicycle, and EV-Charging Supply at the Census-Tract Level

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Abstract—Urban mobility infrastructure—public transit, walkable streets, bicycle facilities, and electric-vehicle (EV) charging—is unevenly distributed, yet existing tools measure each dimension in isolation. We construct a Mobility Equity Score (MES): a transparent, reproducible composite index that integrates all four dimensions at the U.S. Census-tract level, and apply it to three structurally distinct cities (Chicago, Houston, Seattle) using only open data. The MES is strongly spatially clustered in every city (global Moran’s $I = 0.82$ – 0.92) and is positively associated with the share of car-free and renter households. A bivariate negative association with median income is shown to be largely a function of urban form: after controlling for population density and distance to the central business district, and after spatial-regression correction, it attenuates to non-significance in Houston, reverses in Chicago, and persists only in Seattle. Findings are therefore interpreted as associational and consistent with a supply–need mismatch. The pipeline and an interactive dashboard are released open-source.

Index Terms—transportation equity, composite index, spatial analysis, GIS, open data, mobility deserts

I. Introduction

Access to jobs, health care, and education depends on a layered ecosystem of transit, pedestrian, cycling, and—increasingly—EV-charging infrastructure, each with documented equity gaps [1]–[4]. Yet measurement remains siloed: the EPA walkability index, the DOE charger locator, transit GTFS feeds, and OpenStreetMap bicycle data each capture one dimension. No open, reproducible tool integrates all four into a single neighborhood-level equity measure.

This paper contributes: (i) an open, reproducible integration of four mobility dimensions into a tract-level composite; (ii) a transparent methodology whose normalization, weighting, and thresholds are stress-tested; and (iii) an empirical result—that the apparent income gradient in mobility supply is largely an urban-form artifact—obtained through confounder controls and spatial regression. The dashboard is a dissemination layer, not the contribution.

II. Related Work

Single-dimension equity evidence is strong but isolated: transit deserts [1], [5], racialized walkability [2], and severe

TABLE I
The four MES sub-indices.

Sub-index	Metric	Source
Transit	Stop density; AM-peak freq.; span	GTFS
Walkability	Nat. Walkability Index (areal)	EPA SLD
Bicycle	Bike-lane km/km ²	OSM
EV charging	Charger count; dist. to DC-fast	OSM/AFDC

EV-charging gaps for disadvantaged and renter households [3], [6]. Composite-indicator methodology warns that normalization and weighting are value-laden and require sensitivity analysis [7]. Integrated tools exist but trade away tract-level, multi-modal, or open-data properties (e.g., city-scale readiness indices, or spending-based equity tools). The MES fills this gap and, unlike prior work, measures rather than assumes the direction of the supply–demographic relationship.

III. Methodology

A. Study Area and Data

The unit of analysis is the 2023 Census tract for Cook County (Chicago; 1,332 tracts), Harris County (Houston; 1,115), and King County (Seattle; 495). All inputs are open: GTFS (transit), the EPA Smart Location Database (walkability), OpenStreetMap via OSMnx [8] (bicycle and EV charging points), the Census ACS 5-year 2019–2023 (demographics), and TIGER/Line boundaries.

B. Mobility Equity Score

Each of four sub-indices (Table I) is min–max normalized to 0–100 within each city, then combined as an equally weighted mean, $MES = \sum_k w_k s_k^{\text{norm}}$, $\sum_k w_k = 1$. Because normalization is within-city, cross-city levels are not compared; only patterns are. Tracts below the 25th MES percentile are mobility deserts; those below the 25th percentile in ≥ 2 sub-indices are compounded deserts.

C. Analysis

Pearson correlations (with Fisher- z confidence intervals and a Spearman robustness check) relate the MES to

TABLE II

Per-city summary and robustness. Income associations attenuate under urban-form controls and spatial regression (except Seattle); clusters are perfectly stable (ARI); the EV layer is non-decisive.

City	Tracts	Mean MES	Moran's I	MES-income (bivariate)	+ urban-form controls	Cluster ARI
Chicago	1,332	37.5	0.92	− ($p=.002$)	+, weak	1.00
Houston	1,115	31.7	0.83	− ($p<.001$)	n.s. ($p=.64$)	1.00
Seattle	495	36.2	0.82	− ($p<.001$)	− ($p<.001$)	1.00

income, percent non-white, percent zero-vehicle, and percent renter. Global and local Moran's I (queen contiguity, 999 permutations) test spatial clustering. K-means with silhouette-selected k and multi-seed Adjusted Rand Index (ARI) yields a stable typology. Because the MES is spatially autocorrelated, we additionally fit multivariable OLS with urban-form controls (population density, distance to the central business district) and spatial-lag/error models, and we recompute the MES without the EV layer as a robustness check.

IV. Results

A. Distribution and Equity Associations

Walkability is the strongest dimension and dedicated cycling the weakest in all cities. The MES is strongly spatially clustered (Table II: Moran's $I = 0.82$ – 0.92 , all $p < 0.001$), so supply inequity is contiguous and place-based. The MES is positively associated with car-free ($r = 0.41$ – 0.69) and renter ($r = 0.53$ – 0.69) shares, and bivariate negatively with income ($r = -0.08$ to -0.27). Fig. 1 maps the MES and its sub-indices.

B. Spatial Cold Spots, Deserts, and Typology

Local Moran's I (Fig. 2) resolves the clustering into large, statistically significant low-low “cold spots” on each city's periphery and high-high cores, identifying contiguous under-served areas rather than isolated tracts. Because a mobility desert is defined within each city (bottom MES quartile), the $\approx 25\%$ share is definitional; the substantive results are that desert tracts are higher-income peripheries that nonetheless contain large minority populations (52–64% non-white in Chicago/Houston), that the classification is stable under threshold changes (Jaccard 0.80–0.83 across the 20th/25th/30th percentiles), and that compounded deserts (low on ≥ 2 dimensions) concentrate the

disadvantage. K-means recovers interpretable profiles—a well-served multimodal core, a uniformly under-served periphery, and, in Houston, a distinctive “walk-rich, bike-poor” class (537 tracts); silhouettes for the selected k are 0.41–0.48 and the partitions are reproducible across initializations (mean ARI = 1.00). Because k differs by city, similarly named profiles are compared by centroid shape, not label.

C. Confounding, Spatial Dependence, Robustness

Adding density and CBD distance to an OLS of the MES on income changes the income coefficient sharply: it becomes non-significant in Houston, reverses to weakly positive in Chicago, and remains negative and significant only in Seattle. Spatial-lag/error models (error preferred for Chicago, lag for Houston and Seattle by AIC) corroborate this: once spatial dependence is modeled, the income association is negligible except in Seattle, confirming that ordinary significance was overstated. Dropping the OSM-sourced EV layer leaves the findings intact on every axis—the income, renter, and zero-vehicle correlations move by $|\Delta r| \leq 0.05$, the desert set overlaps by a Jaccard of 0.71–0.92, and spatial clustering is essentially unchanged—so the conclusions do not depend on the least-certain dimension.

V. Discussion and Conclusion

What the data demonstrate: mobility supply is strongly spatially clustered, positively associated with car-free and renter shares, and its apparent negative income gradient is largely an urban-form artifact that does not survive density/centrality controls or spatial regression except in Seattle. The composite is well-behaved—related-but-non-redundant sub-indices, stable weighting, reproducible clusters, EV-insensitive conclusions.

What we infer: because supply co-occurs with density (and thus with car-free and renter populations), the equity

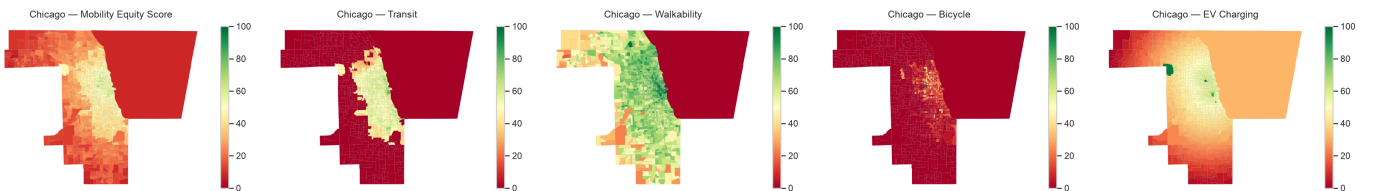


Fig. 1. Chicago MES and its four sub-indices (green = better served, red = worse). Walkability and transit concentrate centrally; dedicated bicycle infrastructure is sparse citywide.

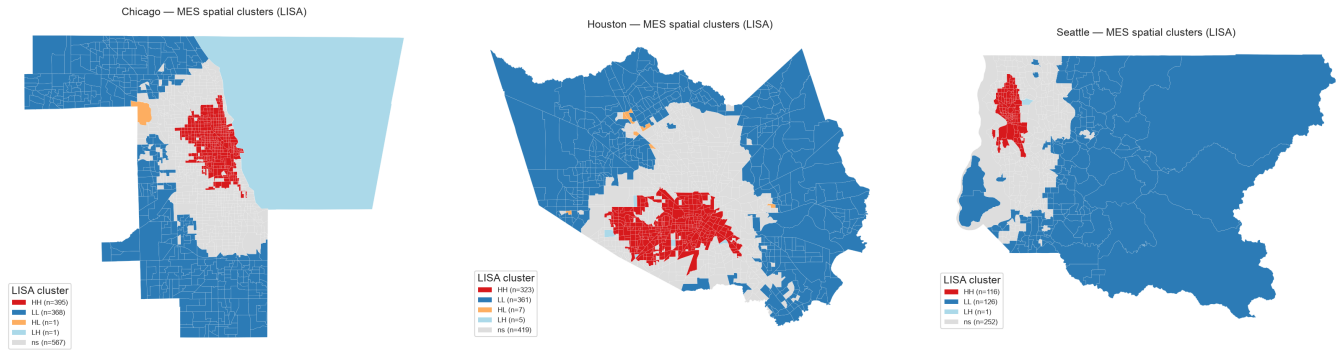


Fig. 2. Local Moran's I (LISA) clusters of the MES (Chicago, Houston, Seattle). Red = high-high (well-served cores); blue = low-low (under-served cold spots).

question is best framed, associationally, as a supply–need mismatch; demand itself is not modeled. Compounded deserts and low-low LISA cold spots locate the tracts where car-free and minority households (52–64% non-white in the Chicago/Houston deserts) plausibly face the widest gap, and are where the framework adds most policy value. In practice this supports neighborhood-level targeting: ranking tracts, separating single- from multi-dimension deficits, and suggesting differentiated interventions such as cycling investment for walk-rich/bike-poor areas and first/last-mile transit for under-served peripheries.

Cross-city reading: because normalization is performed within each city, absolute MES levels are not comparable across cities—a score of 70 in Houston is not equivalent to 70 in Chicago. The robust comparative result is instead that the same qualitative pattern (strong spatial clustering, positive car-free/renter associations, and an income gradient largely explained by urban form) recurs in all three metros, with Seattle the one case where a negative income association survives controls and therefore merits city-specific study.

Limitations are the supply-side scope, the OSM EV proxy, within-city normalization (no cross-city levels), and a three-city cross-section. Future work extends to

more cities, substitutes authoritative AFDC EV and FTA transit service-quality data, and—most directly—develops a demand-adjusted MES that combines infrastructure supply with transportation need so the supply–need relationship can be modeled rather than inferred. The full pipeline and dashboard are open-source, making the method transferable to any city with open data.

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